

1 Solution — GIAM 5.3

Problem 11

1.1 Problem

Give an inductive proof that $\forall n > 3 \in \mathbb{N}, n^3 < 3^n$.

1.2 Answer

$$(k+1)^3 < 3k^3 < 3 \cdot 3^k = 3^{k+1}$$

The first inequality is exactly 5.3 Problem 10, valid for $k \geq 3$; the second is the inductive hypothesis multiplied by 3. With the basis $4^3 = 64 < 81 = 3^4$ this proves the claim for every $n \geq 4$. Verified for every n from 4 to 3000.

1.3 The Proof

Theorem. For every natural number $n > 3$, $n^3 < 3^n$.

Proof. By induction on n , starting at 4.

Basis. At $n = 4$, $4^3 = 64$ and $3^4 = 81$, and $64 < 81$.

Inductive step. Let $k \geq 4$ be a particular but arbitrarily chosen natural number and suppose $k^3 < 3^k$. By 5.3 Problem 10, since $k \geq 3$,

$$3k^2 + 3k + 1 < 2k^3.$$

Adding k^3 to both sides,

$$(k+1)^3 = k^3 + 3k^2 + 3k + 1 < 3k^3.$$

Multiplying the hypothesis by 3 gives $3k^3 < 3 \cdot 3^k = 3^{k+1}$, and chaining the two,

$$(k+1)^3 < 3k^3 < 3^{k+1}.$$

Basis and step together give the result for every $n \geq 4$. ■

If Problem 10 is not to hand, the step also follows directly: for $k \geq 4$, $(k+1)^3/k^3 = (1 + 1/k)^3 \leq (5/4)^3 = 125/64 < 3$.

1.4 The Picture

The table shows where the inequality holds; the curve shows why $n = 3$ is the lone exception.

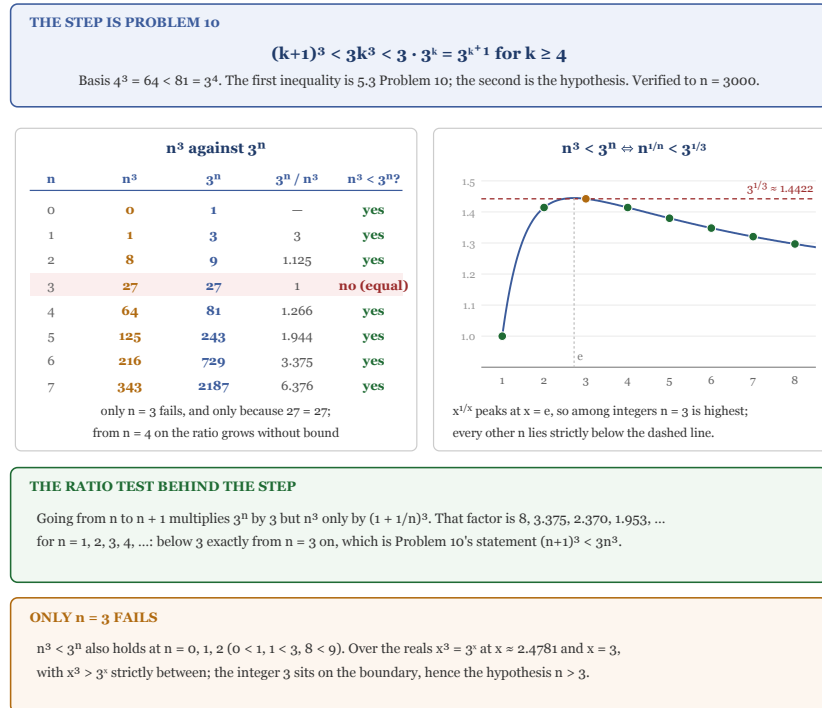


Figure 1.1: A diagram in four parts. The top band gives the step chain, whose first inequality is Problem 10. The left panel tabulates n cubed and three to the n for n from zero to seven; only n equal to three fails, where both are twenty-seven. The right panel graphs x to the power one over x , peaking at e , with only n equal to three reaching the dashed line at the cube root of three. A green band explains the step as a ratio comparison: the factor by which n cubed grows, one plus one over n cubed, drops below three from n equal to three on. An orange band notes the real crossings at about 2.4781 and at 3.

1.5 Remark — Problem 10 Was the Lemma

Problem 10's inequality $3n^2 + 3n + 1 < 2n^3$ looked like an isolated exercise, but it is exactly what this step needs. Since $3n^2 + 3n + 1 = (n+1)^3 - n^3$, it says $(n+1)^3 < 3n^3$: moving from n to $n+1$ multiplies n^3 by less than 3.

That is the general shape of “exponential beats polynomial” proofs. Going from n to $n+1$ multiplies 3^n by 3, but it multiplies n^3 only by the factor $(1 + 1/n)^3$; for $n = 1, 2, 3, 4$ that factor is 8, 3.375, 2.370 and 1.953. Once that factor is below 3, which happens from $n = 3$ on, the exponential's lead can only grow. The basis supplies the lead; the ratio comparison keeps it.

1.6 Remark — Only $n = 3$ Fails

The inequality also holds at $n = 0, 1, 2$: $0 < 1$, $1 < 3$, $8 < 9$. It fails at $n = 3$ alone, where $3^3 = 27 = 3^3$ exactly. Among n from 0 to 99, 3 is the only failure. Over the reals, $x^3 = 3^x$ at $x \approx 2.4781$ and at $x = 3$, with $x^3 > 3^x$ strictly between them.

Taking n -th roots explains the exception. For $n \geq 1$,

$$n^3 < 3^n \iff n^{1/n} < 3^{1/3}.$$

The function $x^{1/x}$ increases up to $x = e$ and decreases after it, because $\ln(x^{1/x}) = (\ln x)/x$ has derivative $(1 - \ln x)/x^2$. So among integers the largest value of $n^{1/n}$ is at $n = 2$ or $n = 3$, and $3^{1/3} \approx 1.4422$ beats $2^{1/2} \approx 1.4142$ because $3^2 = 9 > 8 = 2^3$. Every $n \neq 3$ therefore has $n^{1/n} < 3^{1/3}$ and satisfies the inequality, and $n = 3$ gives equality. The hypothesis $n > 3$ in the problem steps past that single point, and starting at 4 also keeps the step within reach of Problem 10.

1.7 Remark — Exponentials Beat Every Polynomial

The same argument works for any power n^d and any base $a > 1$. The ratio $(1 + 1/n)^d$ tends to 1, so it eventually drops below a , and from then on a^n gains on n^d at every step. What changes is how long the polynomial stays ahead. By exhaustive check, $n^3 < 3^n$ holds for all $n \geq 4$, $n^3 < 2^n$ for all $n \geq 10$, and $n^{10} < 2^n$ for all $n \geq 59$. Each has the same induction as above: a basis at the threshold, and a step that needs $(1 + 1/k)^d < a$.